

Airborne Electromagnetic Characterization of Shallow Permafrost in the Canadian Arctic for Infrastructure Challenges

Introduction

The northern Northwest Territories in the Canadian Arctic contain some of the most ice-rich permafrost landscapes in North America (Kokelj et al., 2023). As climate change drives rising ground temperatures, accelerating permafrost degradation is affecting ecosystems and infrastructure across the region. This project focuses on the Inuvik–Tuktoyaktuk Highway (ITH), a 138 km all-season road in the Northwest Territories that opened in 2017 as the first Canadian highway to reach the Arctic Ocean (Sladen et al., 2022). The ITH traverses continuous permafrost and complex environmental conditions, and plays a critical role in providing access to healthcare, education, and economic opportunities for northern communities. As permafrost degradation increasingly threatens the corridor's stability, sustained monitoring and characterization of subsurface conditions are essential for resilient infrastructure planning. Beyond local impacts, permafrost degradation is also critical because it amplifies Arctic climate feedback (Schuur et al., 2015). To address these challenges, an airborne frequency-domain electromagnetic (AEM) survey was conducted along the ITH in 2024, the longest survey of its kind for continuous permafrost applications. AEM provides a spatially extensive, time-efficient, and cost-effective alternative to traditional methods for regional permafrost characterization. Beyond our AEM-derived subsurface resistivity models, this project validates the results against traditional field measurements and uses surficial geology and remote sensing datasets to support model interpretation and infrastructure hazard assessments. Results from the Hans Creek Bridge (HC) and Zed Creek (ZC) study sites illustrate the project approach and its applications.

Method

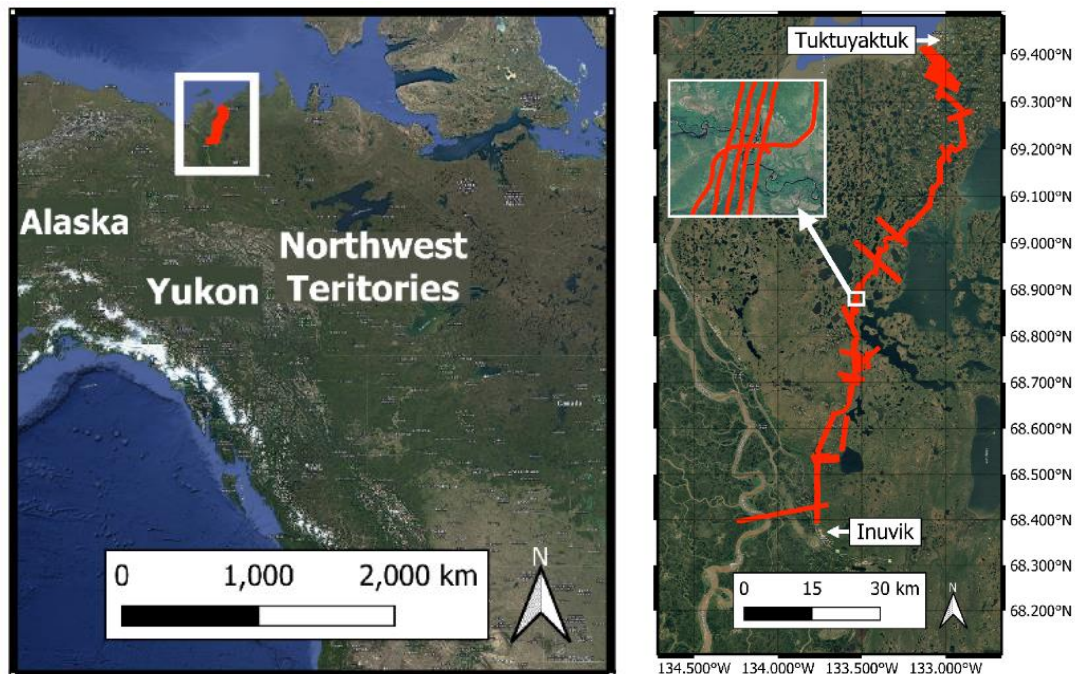


Figure 1: Airborne electromagnetic (AEM) survey lines acquired along the Inuvik–Tuktoyaktuk Highway (ITH) corridor in the Northwest Territories, Canada, highlighting the Hans Creek Bridge and Zed Creek areas of interest.

The AEM survey, in Figure 1, was conducted in August 2024 using the RESOLVE frequency-domain electromagnetic system operated by Xcalibur Multiphysics. The system acquires data at six frequencies: five in coplanar orientation (135 kHz, 40 kHz, 8.2 kHz, 1.8 kHz, and 400 Hz) and one in coaxial orientation (3.3 kHz). A total of 1,142 line-kilometres were flown along the ITH, with 100 m line spacing and an average terrain clearance of 35 m. We inverted this dataset using the open-source SimPEG

framework (Cockett et al., 2015) to recover subsurface resistivity models. Given the survey scale, our inversion workflow generated quasi-2D models for each flight line by stitching together individual 1D inversions. We run first half-space inversions for all soundings along each line, and the mean half-space model was then used as both the starting and reference model for subsequent soundings 1D L2-weighted least-squares inversions. An additional quality-control step improved inversion consistency and reduced overfitting. Additional datasets, including ground-based methods such as Electrical Resistivity Tomography (ERT) profiles, borehole data, and ground temperature measurements, are used to validate and corroborate the AEM-derived models. In addition, remote sensing datasets, including satellite imagery and Interferometric Synthetic Aperture Radar (InSAR) products derived from public datasets, could also be incorporated to provide near-surface constraints.

Hans Creek Bridge

Bridges along the ITH are particularly vulnerable to ground hazards because they concentrate thermal disturbances. This study focuses on the Hans Creek Bridge (HCB) at km 57.3, which is founded on adfreeze pile foundations that require ground temperatures below $-1\text{ }^{\circ}\text{C}$ to maintain structural stability (Siemens & Schetselaar, 2026). Pre-construction data showed that the HCB had the highest ground temperatures at the base of the pier among all ITH bridge locations (Hoeve, 2023), making it a critical site for subsurface monitoring. Several temperature-monitoring boreholes installed by the Northwest Territories Geological Survey (NTGS) exist at the site. However, the dataset contains temporal gaps and limited spatial coverage due to the challenges of traditional borehole monitoring. We use these data here to compare with the AEM-derived results.

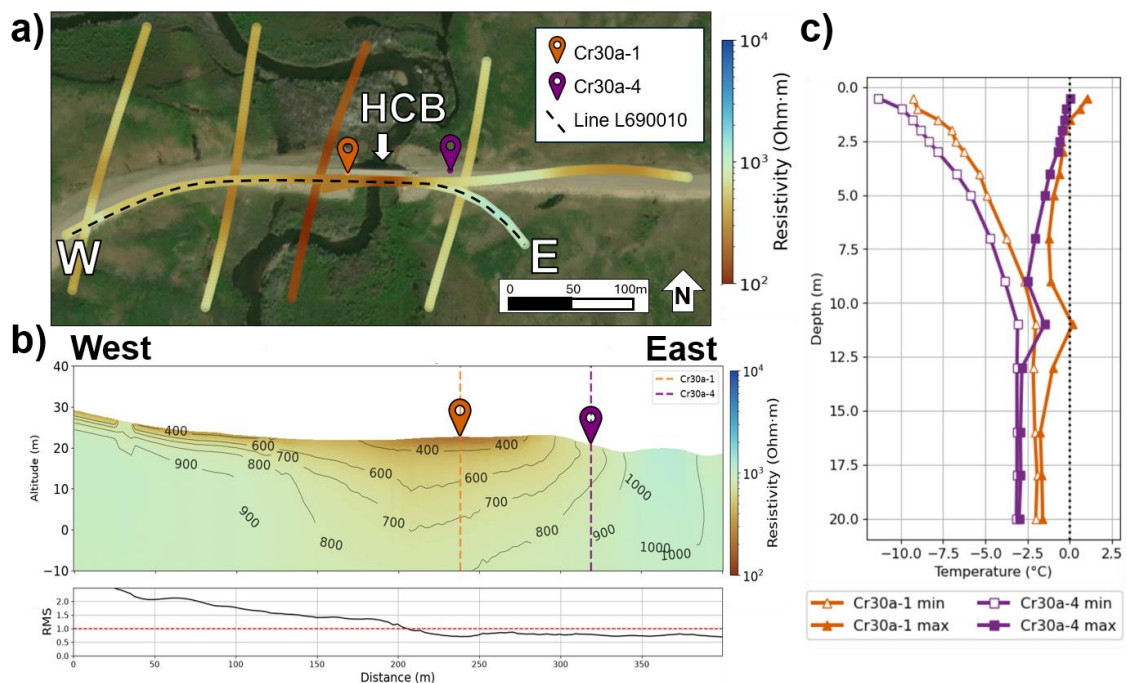


Figure 2: Hans Creek Bridge (HCB) Region: (a) Shallow subsurface resistivity ($\sim 2.25\text{ m}$ depth) derived from AEM flight line inversions; (b) subsurface resistivity cross-section along Line L690010; and (c) borehole temperature profiles from NTGS boreholes Cr30a-1 and Cr30a-3a (2021-12-09 to 2022-06-02).

As shown in Figure 2a, the subsurface resistivity slice at 2.25 m depth reveals a clear contrast, with lower resistivity values on the west side of the bridge compared to the east side. This contrast is also observed in the cross section along Line “L690010” in Figure 2b. The western side of the HCB exhibits lower shallow ($<5\text{ m}$) resistivity values ($\sim 500\text{ }\Omega\cdot\text{m}$), indicative of warmer permafrost conditions with higher unfrozen water content, whereas the eastern side shows higher shallow resistivity values ($\sim 750\text{--}1000\text{ }\Omega\cdot\text{m}$) associated with colder, more ice-rich permafrost. Borehole temperature data from the HCB abutments (Figure 2c) indicate that the western borehole (Cr30a-4) is approximately $1\text{--}2\text{ }^{\circ}\text{C}$ warmer

than the eastern borehole (Cr30a-1), consistent with the resistivity contrasts observed in Figures 2a and 2b. Despite the temporal gap between the borehole records (December 2021–June 2022) and the AEM survey (August 2024), these thermal contrasts are expected to persist and potentially intensify over time, particularly during the summer thaw season. Overall, the HCB results correspondence highlights the value of AEM as a complementary approach to borehole monitoring, especially where priority areas for mitigation are not clearly defined and borehole observations remain spatially limited.

Zed Creek

The Zed Creek (ZC) area is another major region of interest located at km 67–68 along the ITH. Unlike the HCB site, this area has little to no existing public studies or borehole datasets. However, field observations, including ERT profiles, LiDAR surveys, and direct probing, indicate that construction of the ITH altered the local ZC stream system, causing ponding and rerouting of water flow. The dominant surface flow direction is now from south to north, running adjacent to the southeast (SE) side of the ITH (Figure 3a). These hydrological changes have likely contributed to permafrost degradation, subsidence, and washouts affecting both the highway embankment and surrounding ecosystems. Consequently, the region has become a priority area for monitoring due to ongoing infrastructure hazards and stability challenges.

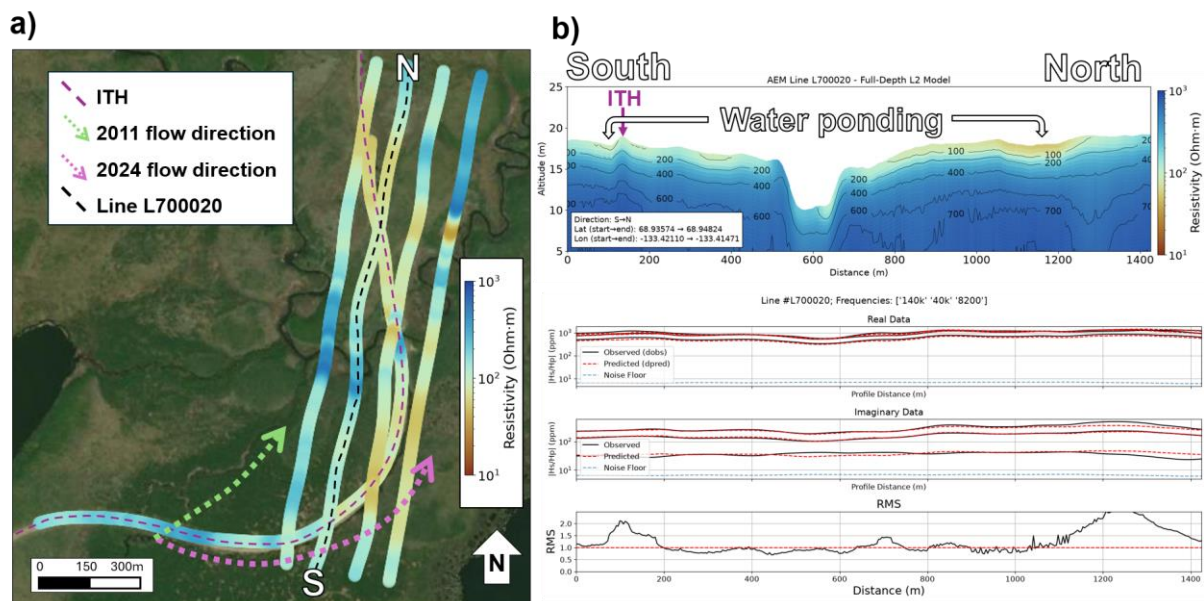


Figure 3: Zed Creek (ZC) Region – (a) Shallow subsurface resistivity (~ 2 m depth) derived from AEM flight line inversions, including illustrative surface flow directions of the Zed Creek stream system before and after ITH construction (green and pink dashed lines); (b) subsurface resistivity cross-section along Line L700020.

As shown in Figure 3, the ZC area exhibits lower resistivity values ($< 500 \Omega \cdot \text{m}$) extending to greater depths (~ 10 m) compared to other areas along the ITH, including the HCB site, suggesting thicker thawed layers and more degraded permafrost. An asymmetry is also observed in the shallowest layer (~ 1 m) in Figure 3a, where the southeast (SE) side of the ITH, area closest to the 2024 surface flow path, exhibits lower resistivity values than the northwest (NW) side (white dashed dividing line). This contrast is also evident in the subsurface model along Line “L700020” (Figure 3b), where the shallowest (< 3 m) on the south side of the highway exhibits lower resistivity values ($\sim 125 \Omega \cdot \text{m}$) than the north side ($\sim 250 \Omega \cdot \text{m}$). Additional smaller localized low-resistivity anomalies are associated with creek crossings and ponded areas, including the “Water ponding” zone east of the ITH, which exhibits very low shallow resistivity values ($\sim 100 \Omega \cdot \text{m}$). These water features and associated sharp transitions are also observed in complementary remote sensing datasets, such as InSAR and satellite imagery. In this case, the surface water presence in the lines informed us to use reference and starting models based on active layer (thawed layer) approximation, improving model consistency in regions with low surficial resistivities and abrupt changes.

Conclusions

The Hans Creek Bridge site provides a strong example of correspondence between the AEM-derived models and existing borehole data, while also mapping regions of shallow resistivity contrast associated with thawing conditions. Complementing this, the Zed Creek area shows good agreement with field observations, documented infrastructure instability, and remote sensing datasets observations. These correspondences are valuable for better constraining and parameterizing the models in areas with sharp surface changes. Overall, the results presented here demonstrate the value of using AEM surveys and derived resistivity models to map structural differences in permafrost terrain, particularly shallow-to-deep resistivity variations at infrastructure scales. Given the strategic importance, scale, geological complexity, and limited accessibility of the ITH corridor, the AEM survey methodology has strong potential as a rapid, large-scale, time-efficient, and cost-effective approach for permafrost characterization. In turn, this method could support infrastructure and hazard planning in remote Arctic regions facing increasing infrastructure instability, as well as growing risks of displacement and relocation driven by thaw-related permafrost degradation and rapidly changing landscape conditions.

Acknowledgements

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